

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 1 of 12

(1) Research Program/Project/Study Title: Post-harvest Behaviour of Organically Grown Cucumber applied with Different Plant Food Supplements	
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Program or Project has component (if no proceed to 2)? ☐ Yes ☐ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☒ Natural/BiologicalSciences ☐ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Exfivesion ☐ Poster	(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☒ CSRDC Crop Science Research Development Center
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(4) Site of implementation: CAPSU Tapaz	
Municipality/City:	Tapaz
Province:	Capiz
Region:	Region VI

(5) Classification <table> <tr> <td>Research:</td> <td>Development:</td> </tr> <tr> <td>☐ Basic</td> <td>☐ Pilot Testing</td> </tr> <tr> <td>☒ Applied</td> <td>☐ Tech. Promotion/Commercialization</td> </tr> </table>	Research:	Development:	☐ Basic	☐ Pilot Testing	☒ Applied	☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☐ Single Authorship/Proponent ☐ Collaborative Research (within campus) ☒ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)
Research:	Development:						
☐ Basic	☐ Pilot Testing						
☒ Applied	☐ Tech. Promotion/Commercialization						

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	
	Document Title:	Document Code	RES- CR-01
		Revision No.	00
		Effective Date	
		Page	Page 2 of 12

(7) Sector/Commodity

Agriculture/Organically Grown vegetables (Radish, kangkong and cucumber)

(8) Discipline/Field of Specialization

Crop Science

(9A) Abstract (This has a maximum of 250 words. It contains 1-2 sentences of rationale, followed by objectives, brief methodology, major findings, and implications of impact of the study.)

Cucumber (*Cucumis sativus* L.) is an economically important crop; it is used for pickling and eaten fresh. Cucumbers are fruit, and approximately 95% water. They are harvested while still immature, before the seeds enlarge and harden. It is difficult to store. Cold temperatures cause chilling injury, while warmer temperatures increase rots and yellowing. There are several studies focusing on the postharvest treatments in prolonging the shelflife of cucumber produce synthetically. However, there are limited information on the postharvest behaviour of organically grown cucumber. Hence, this study was conducted to determine the postharvest behaviour of organically grown cucumber applied with different plant food supplements.

The study was conducted using a single factor CRD. Treatments were corresponding to the different plant food supplements namely: T₁ – Vermicast, T₂ – Vermitea and T₃ – Ordinary Compost. Each treatment were replicated 3 times. The data were analyzed using the Analysis of Variance and F-test for Factorial CRD. Differences among treatment means was determined using Least Significant Differences (LSD), interpreted at 5% and 1% levels of significance. The data shows that cucumber applied with pre-harvest treatment of plant food supplements significantly influenced the Cumulative weightloss and color Index vermicast as the highest in the 12th day of storage of cucumber fruits. Moreover, cucumber applied with ordinary compost significantly influenced the Final TSS and observed with an increasing trend. However, cucumber applied with different plant food supplements did not significantly influenced the Visual quality Rating, Firmness and pH of the cucumber fruits.

(9B) Keywords: *Firmness, Organic, postharvest behaviour and total soluble solids*

(10) Introduction

Cucumber (*Cucumis sativus* L.) is an economically important crop; it is used for pickling and eaten fresh. Cucumber fruits are a good source of antioxidants, magnesium, and vitamin C and are rich in dietary fiber (Shi et al., 2015). However, cucumber fruits have a short storage duration, limited to less than 14 days due to loss of weight and firmness, discoloration, and fungal infections (Bahnasawy and Khater, 2014; Martin-Belloso and Fortuny, 2011). The popularity of cucumber has been on the increase over the years due to its nutritional and economic benefits.

Cucumbers are fruit, and approximately 95% water. They are harvested while still immature, before the seeds enlarge and harden. In some varieties the core flesh develops a translucent appearance at maturity. Cucumbers are extremely variable. Gentle handling of these commodity is essential to prevent cuts, scratches or bruises, as these are likely to develop into rots. Cucumbers should never be packed wet as this is likely to increase disease. Washing thin skinned varieties is not recommended.

The increase in soil productivity is one of the major key factors, attributing to substantial increase in agricultural production to fulfill the increase in the human population. In order to achieve higher yields and quality, soil health is a critical factor. Therefore, organic practice must be done. Mineral fertilizers considered a major source of plant nutrition, but the excessive use of mineral fertilizers represent the major cost in plant production and creates pollution of agro-eco system as well as deterioration of soil fertility (Singh and Ryan, 2015).

Cucumber is difficult to store. Cold temperatures cause chilling injury, while warmer temperatures increase rots and yellowing. Chilling sensitivity is strongly affected by variety and the

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 3 of 12

growing conditions leading up to harvest. There are several studies focusing on the postharvest treatments in prolonging the shelflife of cucumber produce synthetically. However, there are limited information on the postharvest behaviour of organically grown cucumber. Hence, this study was conducted to determined the postharvest behaviour of organically grown cucumber applied with different plant food supplements.

(11) Materials and Methods or Methodology (This includes the materials, methods,duration & location of the study. Materials and Methods are headings used in Natural Science researches and methodology is Social Sciences. Presentation of materials may not necessarily be in separate heading but may be incorporated in the specific methods used. Justify paragraphs with main headings in caps.)

Experimental treatments and design. The study was conducted using a single factor CRD. Treatments were corresponding to the different plant food supplements namely: T₁ – Vermicast, T₂ – Vermitea and T₃ – Ordinary Compost. Each treatment were replicated 3 times. The experimental lay-out was shown in Figure 1.0.). Each treatment per replication was represented by 3 samples.

Preparation of Expiremental samples. Experimental samples was obtained from organic area of Capiz State University, Burias Campus. This was done by selecting samples with the same sizes and color of the cucumber.

Replication 1	Replication 2	Replication 3
T2	T3	T1
T1	T2	T3
T1	T3	T2

Figure 1.0 The experimental lay-out of the study arranged in Complete Randomized Design.

Legend:

Treatments

T1 – Vermicast

T2 – Vermitea

T3 – Ordinary Compost

The data gathered were:

Post-Harvest Qualities

Physical Charectersistics

The physical appearance of fruit was assessed every other day using the following visual quality rating (VQR) of Acedo (2004).

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 4 of 12

VQR was evaluated on a 9-1 scale, 9 as Excellent, field fresh or no defect, 7 as good, defects minor, 5 as Fair, defects moderate, 3 as Poor, defects serious, limit of marketability, 2 as limit of edibility and 1 as Non-edible under usual condition.

Cumulative Weight Loss (%) was taken from the initial weight and the weight of the fruits was measured every other day and subtracted to the initial weight/initial weight x 100.

Firmness was determined using the penetrometer device to analyze the sensory firmness index which was gathered from initial to final.

Fruit shriveling was determined using the Shriveling Index (SI) scale of 1 (No shriveling), 2 (A light shriveling with 1-25% fruit surface affected), 3 (Moderate shriveling at 26-50% fruit surface affected), 4 (Severe shriveling with more than 50% fruit surface affected). With regards to peel color development.

Color Index. Color associated with ripening was determined at each observation period using the color index (CI) scale, 1 (green), 2 (breaker, define a break in color but less than 10% of the surface), 3 (turning, more green than yellow, pink, orange, or red), 4 (more yellow, orange, pink, or red than green color), 5 (yellow, orange, pink, or red with a trace of green), 6 (full yellow, orange or red).

The postharvest life from the sample fruits was evaluated by exposing at room temperature and was recorded until the time when the VQR of 3 (poor, defects serious, limit of marketability) was attained.

Cemical charecteristics

Potential Hydrogen (pH) was measured using pH meter from the extracted juice or aliquot of organically grown Cucumber. Total soluble solids (TSS) were analyzed with the use of a hand-held refractometer.

Statistical Tools and Analysis

The data were analyzed using the Analysis of Variance and F-test for Factorial CRD. Differences among treatment means was determined using Least Significant Differences (LSD), interpreted at 5% and 1% levels of significance.

(12) Results and Discussions

Cumulative Weight Loss (%)

Cumulative weight loss of cucumber is presented in Figure 1. The result showed a consistent increase in a weight loss of the fruits from second (2) day to thirty two (32) days of storage, regardless of the different plant food supplement with no significant variation among treatments except for the twelve (12) day of storage. A slope increase on weight loss was observed on the 12 day of cucumber applied with vermicast with a mean of 12.03% which was followed by T2- vermitea and T3- ordinary compost with a mean of 9.09% and 8.65%, respectively.

Weight loss in cucumber is attributed to the function of the different metabolic processes such as respiration, oxidative processes Wilson (2017) and moisture loss (Kapsiya et al 2015).

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 5 of 12

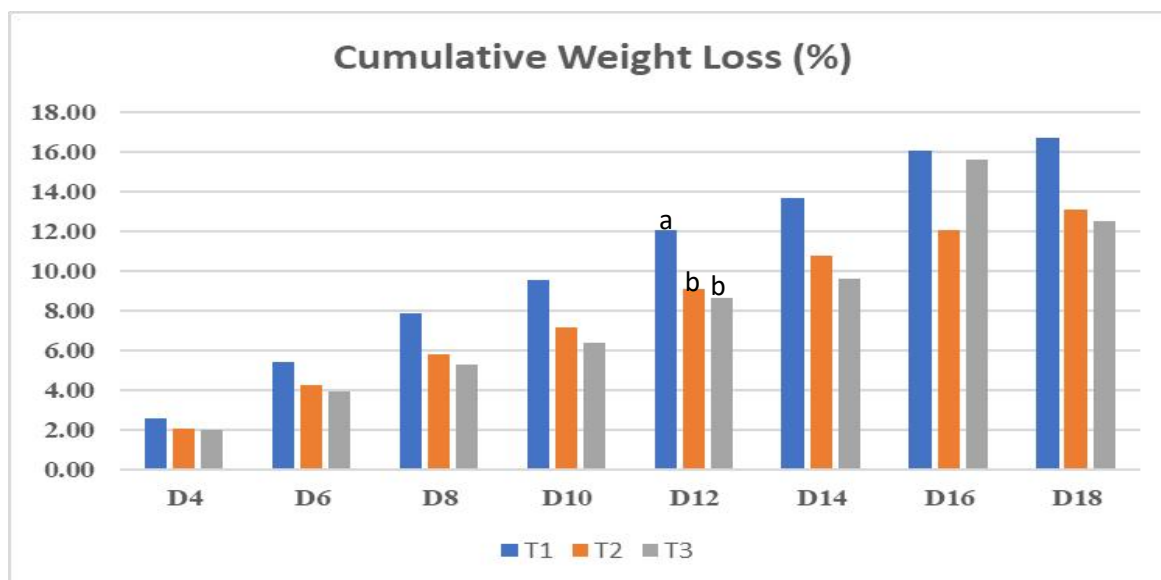


Figure 1.0. Cumulative weight loss of cucumber applied with different plant food supplements.

Visual Quality Rating

Juxtaposing on the results of visual quality rating (VQR) of cucumber, the results did not significantly affect by the different plant food supplements (Figure 2.0). All the fruits had a visual quality of 9, which was described as excellent, field fresh and without defects on the fourth (4th) day of storage. A declined on VQR was observed on the sixth (6th) day of storage until the eighteenth (18th) day of storage. But, there is no significant difference among treatments that were observed during the evaluation of fruits. Nonetheless, it was noticeable that cucumber fruits with a visual quality rating of two were first attained after sixteenth (16th) day of storage from plants applied with vermitea with a comparable effect to vermicast and ordinary compost.

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	
			Page	Page 6 of 12

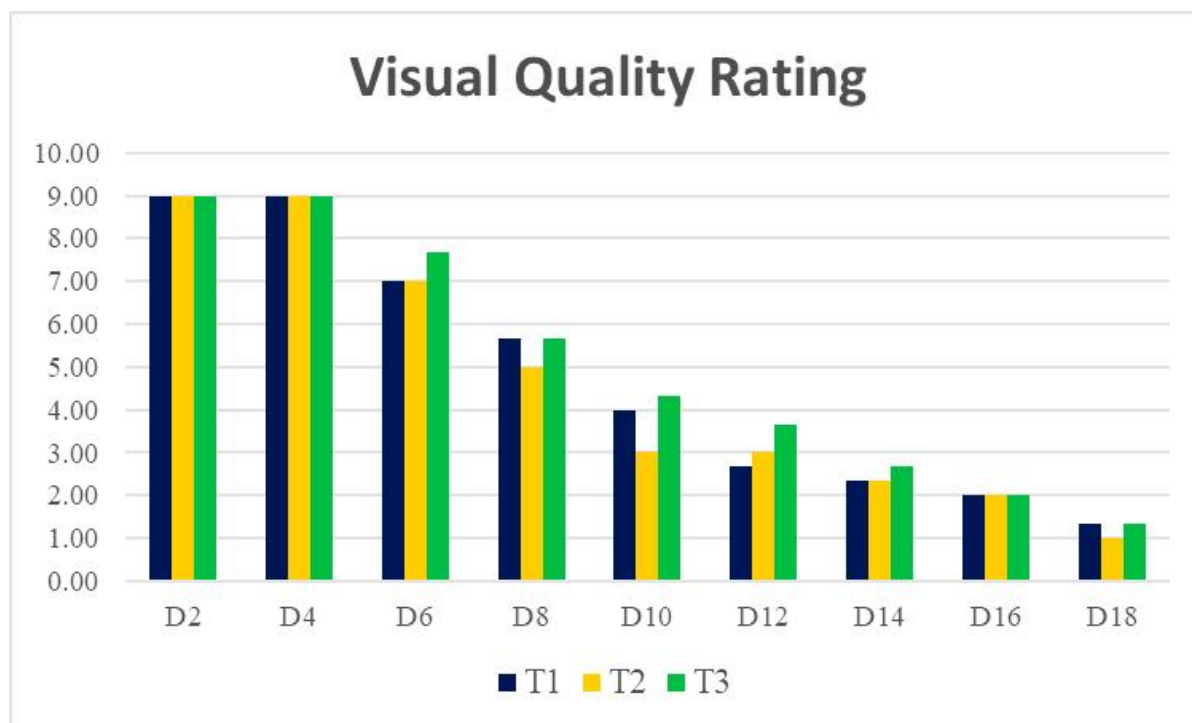
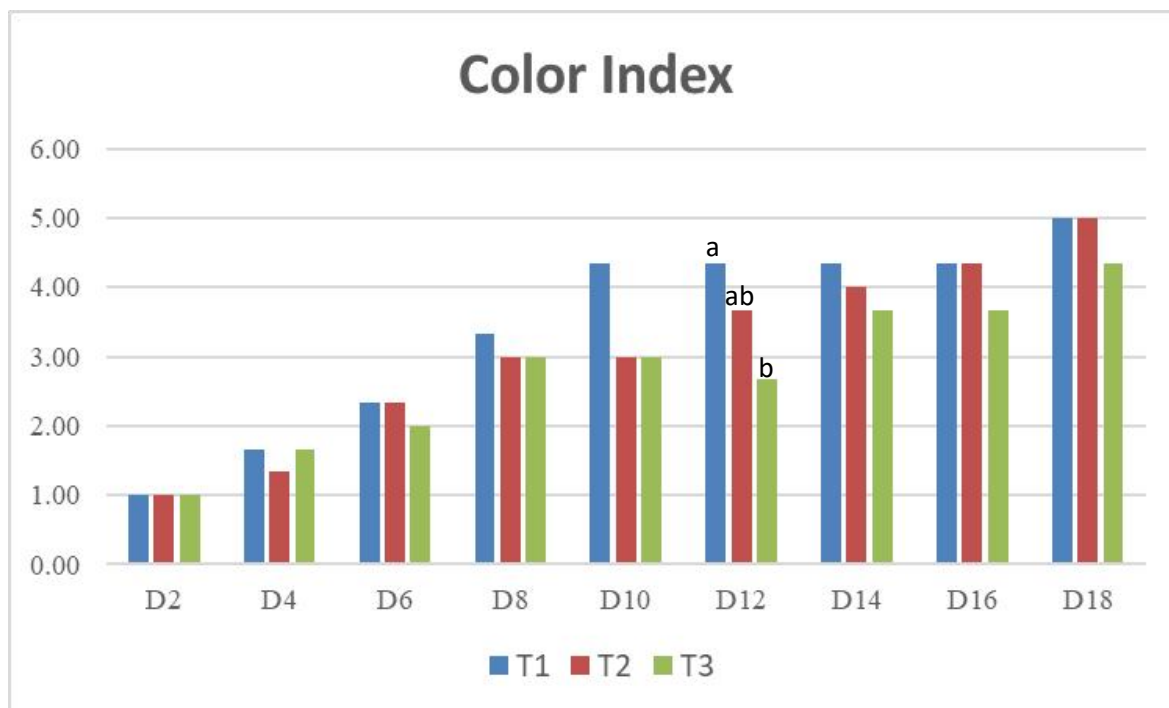


Figure 2.0. Visual Quality Rating (VQR) of cucumber applied with different plant food supplements.

Color Index

The color index of cucumber fruits was not significantly influenced by the different plant food supplements except for the twelve (12) day of storage (Fig. 3). The results showed that there is no significant variation among treatments in storage day 2, 4, 6, 8, 10, 14, 16 and 18, respectively. However, a significant variation was observed on the twelve (12) day of storage. It was observed that Treatment 1 (Vermicast) has the highest color changes among treatments in terms of numerical value which was comparable to Treatment 2 (Vermitea) and the least color changes was observed in treatment 3 (ordinary compost). Color changes in fruits are results of the disappearance of chlorophyll. The chloroplast had undergone an ultrastructural and functional transformation into chromoplast which results in unmasking the carotenoids (yellow/red color). Anthocyanin also contributes to colors in some ripe fruits and vegetables Kasim and Kasim (2014).

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 7 of 12



Initial and Final Firmness

The initial and final firmness rating of bell cucumber applied with different plant food supplements were presented in (Table 1). The results revealed that there is no significant difference among treatments. However, there was a rapid decrease in firmness from initial to final results. Higher initial and final firmness was from harvested fruits applied with ordinary compost (7.25 and 6.04) followed by fruits applied with vermicast (5.98 and 5.67) and the softest fruits were observed from harvested fruits applied with vermitea with a mean of (5.52 and 5.23) . During fruit ripening, one of the most notable changes is softening which is related to biochemical processes in the cell wall (Pinheiro et al 2013). Bertin and Genard (2018) reported that the membrane and cell wall breakdown occurs due to less turgor and enzyme orchestrated cell wall losing. As cited by Brummell (2006), the key factors associating with fruit softening are the depolymerization and degradation of cell components and the level of polygalacturonase activity has been positively correlated with fruit ripening and softening in banana. Softening is also brought about by pectin degradation by pectin methylesterase.

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 8 of 12

Table 1.0. Initial and final firmness of cucumber applied with different plant food supplements

TREATMENTS	Initial Firmness	Final Firmness
T1- Vermicast	5.98	5.67
T2- Vermitea	5.52	5.23
T3- Ordinary Compost	7.25	6.04
P-value	0.1203 ^{ns}	0.2096 ^{ns}
CV(%)	14.11	8.69

Initial pH and Final pH

Application of plant food supplements at pre-harvest did not significantly affect the pH of cucumber fruits (Table 2). Results showed that there was an increase in the pH of cucumber from initial up to the final day of storage. The highest initial pH was observed from Treatment 3 (ordinary compost) and the lowest pH was obtained from fruits applied with vermicast (T₁) with a comparable effect to fruits applied with vermitea (T₂). Moreover, the highest final pH was obtained from fruits applied with the pre-harvest application of vermicast with a mean of 5.88 which was found comparable to fruits applied with vermitea and the lowest was observed from Treatment 3- ordinary compost (7.33).

Initial TSS and Final TSS

Total soluble solids (TSS) of cucumber as influenced by different plant food supplements were presented in (Table 2). Results revealed that there was a significant difference in initial and final TSS of cucumber. Fruits applied with vermitea obtained the highest initial TSS with a mean of 1.39Brix this was followed by fruits applied with vermicast and the least initial TSS was observed from fruits applied with vermitea with a mean of 1.34 and 1.33brix, respectively. However, the highest final TSS was observe from fruits applied with ordinary compost which was found comparable with treatment 1 vermicast and the lowest final TSS was observed from treatment 2 vermitea (1.49, 1.32, and 1.03Brix, respectively).

An increasing trend in TSS was observed from treatment 3- ordinary compost with a mean of (1.33 – 1.49Brix). Increased in TSS was due to the increase of water loss (weight loss) during storage (Bayoumi 2008). Akhtar et al (2010) reported that increasing TSS content is due to the concentrated juice content of the fruits. However, decreasing TSS was observed from treatment 1 and 2 (vermicast and vermitea) with a mean of 1.34-1.32 and 1.39-1.03Brix, respectively.

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 9 of 12

Table 2.0. Initial pH, Final pH, Initial TSS and Final TSS of cucumber applied with different plant food supplement.

TREATMENTS	Initial pH	Final pH	Initial TSS(Brix)	Final TSS(Brix)
T1- Vermicast	5.88	7.55	1.34 _b	1.32 _{ab}
T2- Vermitea	5.88	7.55	1.39 _a	1.03 _b
T3- Ordinary Compost	5.92	7.33	1.33 _b	1.49 _a
P-value	0.6419 ^{ns}	0.7786 ^{ns}	0.0210*	0.0307*
CV(%)	1.07	9.29	1.54	12.16

(13) Conclusions and Recommendations

Conclusions

Based on the results obtained, the following conclusions were drawn:

1. Application of different plant food supplement significantly influenced the Cumulative weightloss, Color index, initial and final TSS on the 12th day of storage of Cucumber applied at pre-harvest treatment
2. Visual quality rating, initial and final firmness and initial and final pH were not affected by different plant food supplements applied at pre-harvest treatments of cucumber

Recommendations

Based on the findings of the study, the following recommendations were forwarded:

1. Use ordinary compost as preharvest treatment for increasing TSS in cucumber fruits.
2. Conduct similar studies using other plant food supplements

(14) Output (Indicate the specific products, processes or services which the project has produced; quantify when possible.) Mushroom

(15) Technology Transfer Activity Conducted (Indicate the specific products, activity design, processes or services transferred as exfivesion activities)

(16) Economic Analysis (This should present the direct cost of project implementation, expenses in the production of technology, revenues generated from technology produced, return on investment (ROI) and payback period.

Sample Economic Analysis

REVENUE

Sales

DIRECT COST

Direct Charges- Wages & Labor

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 10 of 12

Direct Charges - Materials & Supplies

Direct Charges - Depreciation

Direct Charges - Repair/Mainfiveance &
Fuel/Oil

COST OF TECHNOLOGY PRODUCED

GROSS PROFIT

PRODUCTION EXPENSES

Salaries & Wages

Lights and Water

Telecommunications

Transportation

Office Supplies/Supplies

Permits & Licenses

Rentals and Expenses

Miscellaneous

Advertising & Promotion

Freight Expense

Total Production Cost

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 11 of 12

NET INCOME

=====

Net Income

ROI (%) = _____

Total Project Cost

Total Project Cost

Payback Period (Years) = _____

Average Annual Profit

(17) Acknowledgments (Include only important persons and institutions with involvement in funding and conduct of the research. Justify paragraphs with main headings in caps.)

We would like to express our deep appreciation and gratitude first and foremost to the Almighty God for His unending love. To the University president, Dr. Editha C. Alfon, the Vice President for Research, Development and Extension, Eng. Efren L. Linan. To Dr. Salvacio J. Legaspi Center Director, Crop Science, for their Guidance and valuable support, to the Campus Administrator of CapSU Tapaz, Dr. Arlene C. Gonzalez, for her moral support.

(18) References (It should include only works mentioned in the text and should be arranged alphabetically. Citations shall not be less than 13 entries. This must follow the APA latest edition format, in hanging indentation with 12-point space in between.)

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	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	
			Page	Page 12 of 12

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